PHYSICS

SECTION-1

1) Two identical balls moving with a speed of 6 m/s in opposite direction, collides. If for collision, $e = \frac{1}{3}$; the speed of each ball after collision will be :-

(1) 2 m/s
(2) 4 m/s
(3) 6 m/s
(4) 1 m/s

2) **Assertion :** In an unbiased p-n junction, holes diffuse from the p-region to n-region.
Reason : Hole concentration in p-region is more as compared to n-region.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

3) Match the columns :

	Column-I		**Column-II**
i.	Sun	p.	Nuclear fission
ii.	Nuclear reactors	q.	Nuclear fusion
iii.	Total nuclear binding energy in a process is increased	r.	Energy is released
iv.	Total nuclear binding energy in a process is decreased	s.	Energy is absorbed

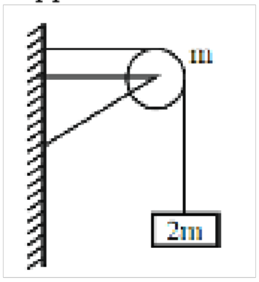
Now, select the correct option from the codes given below :
(1) i-q,r, ii-p,r, iii-s, iv-q
(2) i-p,r, ii-q,r, iii-r, iv-s
(3) i-q,r, ii-p,r, iii-p,q,r, iv-s
(4) i-p,r ii-q,r, iii-p, iv-q

4) When work is done on a body by an external force, its

(1) Only kinetic energy increases

(2) Only potential energy increases
(3) Both kinetic and potential energies may increase
(4) Sum of kinetic and potential energies remains constant

5) An ideal string passing over a clamped pulley of mass m supports a block of mass 2m as shown in the figure. The force on the pulley by the clamp is given as



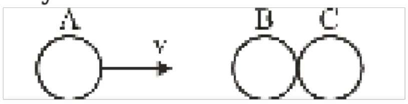
(1) $2\sqrt{2}mg$
(2) 2mg
(3) $\sqrt{13}mg$
(4) Zero

6) On increasing temperature of a p-n junction diode

(1) forward current increases while reverse current decreases
(2) forward current decreases while reverse current increases
(3) forward current remains constant while reverse current increases
(4) forward current increases while reverse current remains constant

7) As shown in figure A, B and C are identical balls. B and C are at rest and the ball A moving with velocity v collides elastically with ball B. Then after collision :

(all collision are elastic)



(1) All the three balls move with velocity v/2
(2) A comes to rest and (B + C) moves with velocity $v/\sqrt{2}$
(3) A moves with velocity v and (B + C) moves with velocity v
(4) A and B come to rest and C moves with velocity v

8) When P–N junction diode is forward biased, then:-

(1) the depletion region is reduced and barrier height is increased
(2) the depletion region is widened and barrier height is reduced
(3) both the depletion region and barrier height are reduced
(4) both the depletion region and barrier height are increased

9) A particle of mass m is driven by a machine that delivers a constant power k watts. If the particle starts from rest the force on the particle at time t is :

(1) $\sqrt{mk}t^{-1/2}$

(2) $\sqrt{2mkt}^{-1/2}$

(3) $\frac{1}{2}\sqrt{mkt}^{-1/2}$

(4) $\sqrt{\frac{mk}{2}}t^{-1/2}$

10) A balloon with mass 'm' is descending down with an acceleration 'a' (where a < g). How much mass should be removed from it so that it starts moving up with an acceleration 'a' ? (Assume that it's volume does not change)

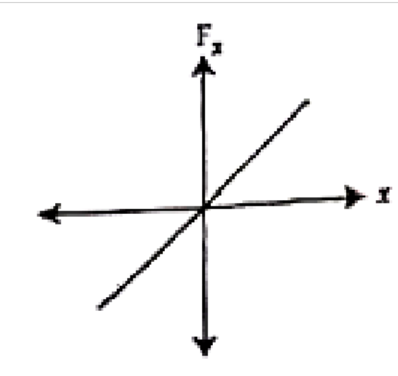
(1) $\frac{2ma}{g+a}$

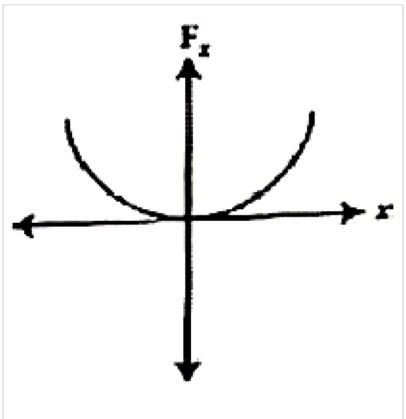
(2) $\frac{2ma}{g-a}$

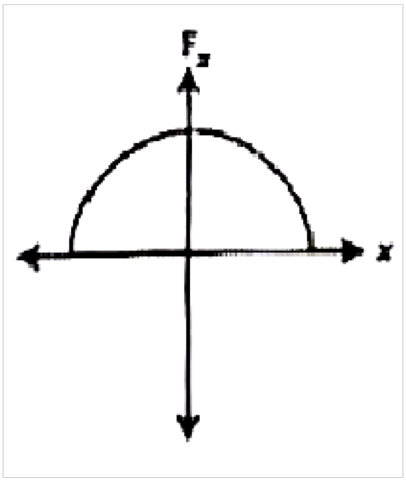
(3) $\frac{ma}{g+a}$

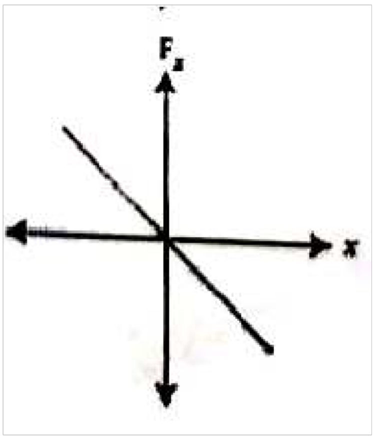
(4) $\frac{ma}{g-a}$

11) The restoring force of a spring with a block attached to the free end of the spring is respresented by :

(1) 

(2) 

(3) 

(4) 

12) **Assertion :** Same force applied for the same time causes the same change in momentum for different bodies.
Reason : The total momentum of an isolated system of interacting bodies remains conserved.

(1) Assertion is correct, reason is correct; reason is a correct explanation for assertion.
(2) Assertion is correct, reason is correct; reason is not a correct explanation for assertion
(3) Assertion is correct, reason is incorrect
(4) Assertion is incorrect, reason is correct.

13) A particle loses 25% of its kinetic energy during collision with another identical particle at rest. The coefficient of restitution will be :

(1) 0.25
(2) $\sqrt{2}$
(3) $\frac{1}{\sqrt{2}}$
(4) 0.5

14) A chain of mass 2 kg is placed on a smooth table with 1/4th of its length hanging over the edge. If the length of chain is 4 m then the work done in pulling the hanging portion of the chain back to the surface of the table is

(1) 5 J
(2) 2.5 J

(3) 2 J

(4) 4 J

15)

The current density (J) for an intrinsic semiconductor is given by which of the following equations :-
[where μ_e = electron mobility
μ_h = hole mobility
n_1 = carrier concentration

E = applied electric field]

(1) $J = n_1 . e . (\mu_e + \mu_h) . E$

(2) $J = n_1 . e . (\mu_h - \mu_e) . E$

(3) $J = \frac{n_1 \cdot e \cdot (\mu_e - \mu_h)}{E}$

(4) $J = \frac{E}{n_1 \cdot e \cdot (\mu_e + \mu_h)}$

16) A coin is dropped in a lift. It takes time t_1 to reach the floor when lift is stationary. It takes time t_2 when lift is moving up with constant acceleration. Then

(1) $t_1 > t_2$

(2) $t_2 \geq t_1$

(3) $t_1 = t_2$

(4) Depends on mass of coin

17) In semiconductors at room temperature the :-

(1) valence band is partially empty and the conduction band is partially filled

(2) valence band is completely filled and the conduction band is partially empty

(3) valence band is completely filled

(4) conduction band is completely empty

18) A train is moving with velocity 20 m/sec. On this dust is falling at the rate of 50 kg / minute. The extra force required to move this train with constant velocity will be :-

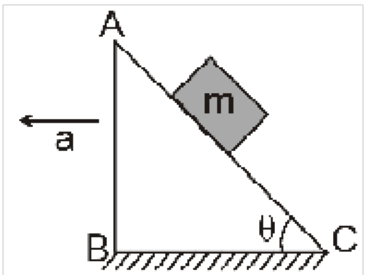
(1) 16.66 N

(2) 1000 N

(3) 166.6 N

(4) 1200 N

19) A block of mass m resting on a wedge of angle θ as shown in the figure. The wedge is given an acceleration a towards left. What is the minimum value of a due to external agent so that the mass m

falls freely? 

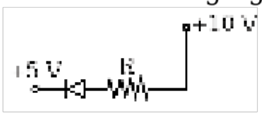
(1) g

(2) g cosθ

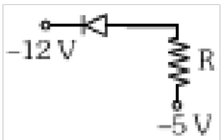
(3) g cotθ

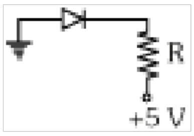
(4) g tanθ

20) In the following figure, the diodes which are forward biased, are :-

(a) 

(b) 

(c) 

(d) 

(1) (a), (b) and (d)

(2) (c) only

(3) (a) and (c)

(4) (b) and (d)

21) Binding energy of $^2He^4$ and $_3Li^7$ are 27.37 MeV and 39 MeV respectively. Which of the two nuclei is more stable ?

(1) He^4

(2) Li^7

(3) Equally

(4) Can't decide

22) A block of mass m is lying on an inclined plane. The coefficient of friction between the plane and the block is μ. The force (F_1) required to move the block up the inclined plane will be–

(1) mg sinθ + μmg cosθ

(2) mg cosθ – μmg sinθ

(3) mg sinθ – μmg cosθ

(4) mg cosθ + μmg sinθ

23) If by successive disintegration of $_{92}U^{238}$, the final product obtained is $_{82}Pb^{206}$, then how many number of α and β particles are emitted

(1) 8 and 6

(2) 6 and 9

(3) 12 and 6

(4) 8 and 12

24) A locomotive of mass m starts moving so that its velocity varies according to the law $v = k\sqrt{S}$ where k is constant and S is the distance covered. Find the total work performed by all the forces which are acting on the locomotive during the first t seconds after the beginning of motion.

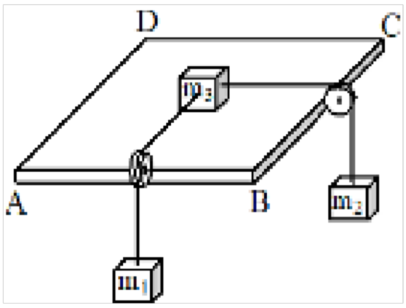
(1) $W = \frac{1}{8} mk^4t^2$

(2) $W = \frac{1}{4} m^2k^4t^2$

(3) $W = \frac{1}{4}mk^4t^4$

(4) $W = \frac{1}{8} mk^4t^4$

25) Three blocks are arranged as shown in which ABCD is a horizontal plane. Strings are massless and both pulley stands vertical while the strings connecting blocks m_1 and m_2 are also vertical and are perpendicular to faces AB and BC which are mutually perpendicular to each other. If m_1 and m_2 are 3 kg and 4 kg respectively. Coefficient of friction between the block m_3 = 10 kg and surface is μ = 0.6 then, frictional force on m_3 is -



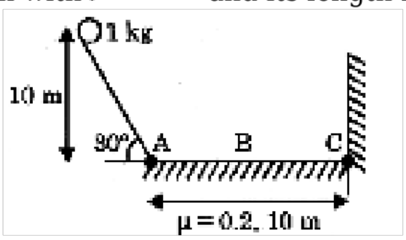
(1) 30 N

(2) 40 N

(3) 50 N

(4) 60 N

26) A ball at rest is released from height 10 m at inclined at an angle 30° with horizontal. Horizontal track is rough with $\mu = 0.2$. and its length is 10 m, then total distance travelled by ball before it stops will be



(1) 86 m

(2) 50 m

(3) 76 m

(4) 102 m

27) A person is standing in an elevator. In which situation he finds his weight less than actual when:-

(1) The elevator moves upward with constant acceleration

(2) The elevator moves downward with constant acceleration.

(3) The elevator moves upward with uniform velocity

(4) The elevator moves downward with uniform velocity

28) A particle moves in a straight line with retardation proportional to its displacement. Its loss of kinetic energy for any displacement x is
proportional to

(1) x^2

(2) e^x

(3) x

(4) $\log_e x$

29) In position A kinetic energy of a particle is 60 J and potential energy is –20 J. In position B, kinetic energy is 100 J and potential energy is 40J. Then, in moving the particle from A to B:

(1) Work done by conservative force is 50 J

(2) Work done by external force is 40 J

(3) Net work done by all the forces is 40 J

(4) Net work done by all the forces is 100

30) A particle of mass m moving with velocity u strikes a stationary particle of mass 2m and sticks to it. The speed of the system will be

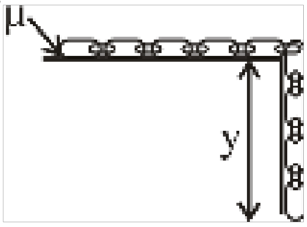
(1) u/2

(2) 2u

(3) u/3

(4) 3u

31) A chain of M, L is placed on table as shown find the maximum value of hanging part so that chain remain at rest :-



(1) $y_{max} = \frac{L}{1+\mu}$

(2) $y_{max} = \frac{1+L}{\mu}$

(3) $y_{max} = \frac{\mu L}{1+\mu}$

(4) None of the above

32) Match the column

	Column I		Column II
a	Limiting friction	p	Has magnitude $\mu_k N$
b	Static friction	q	Maximum value of static friction
c	Kinetic friction	r	Is a self adjusting force

(1) a→p, b→q, c→r

(2) a→r, b→p, c→q

(3) a→q, b→r, c→p

(4) a→q, b→p, c→r

33) Which of the following statements are incorrect:-
(a) the rest mass of a stable nucleus is less than the sum of the rest masses of its separated nucleons
(b) the rest mass of a stable nucleus is greater than the sum of the rest masses of its separated nucleons
(c) in nuclear fusion, energy is released by fusing two nuclei of medium mass (approximately 100)
(d) in nuclear fission, energy is released by fragmentation of a very heavy nucleus

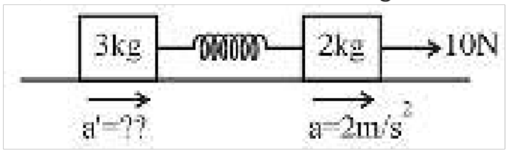
(1) a, b

(2) b, c

(3) c, d

(4) a, d

34) Find acceleration of 3 Kg mass when acceleration of 2 Kg mass is $2 m/s^2$ as shown in figure :-



(1) $3\ m/s^2$

(2) $0.5\ m/s^2$

(3) $2\ m/s^2$

(4) 0

35) A balloon has 5 g of air. A small hole is pierced into it. The air escapes at a uniform rate with a velocity of $4\ cms^{-1}$. If the balloon shrinks completely in 2.5 seconds, then the average force acting on the balloon is

(1) 2 dyne

(2) 50 dyne

(3) 8 dyne

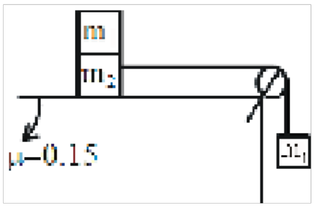
(4) 8 N

SECTION-2

1) The binding energy of α-particle is (if m_p = 1.00785 u, m_n = 1.00866 u and m_α = 4.00274 u)

(1) 56.42 MeV
(2) 2.82 1 MeV
(3) 28.21 MeV
(4) 32.4 MeV

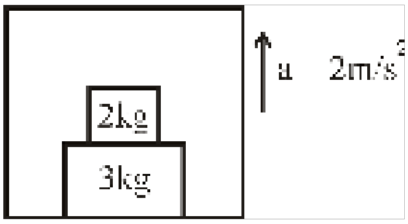
2) Two masses m_1 = 10 Kg and m_2 = 20 Kg connected by a inextensible string over a frictionless pulley, are moving in the figure. The coefficient of friction of horizontal surface is 0.15. The minimum mass m that should be put on top of m_2 to stop the motion is :



(1) $\frac{260}{3}$Kg
(2) $\frac{140}{3}$Kg
(3) $\frac{100}{3}$Kg
(4) 100 Kg

3) Given below are two statements : one is labelled as Assertion (A) and other is labelled as Reason (R).
Assertion (A) : Neutrons penetrate matter more readily as compared to protons.
Reason (R) : Neutrons are slightly more massive than protons.

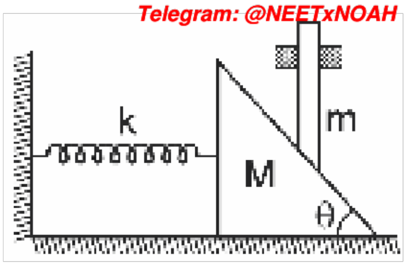
(1) If both (A) and (R) are true but (R) is not the correct explanation of the (A).
(2) If (A) is true but (R) is false.
(3) If both (A) and (R) false.
(4) If both (A) and (R) are true and (R) is the correct explanation of the (A).

4)


Find out normal reaction acting between 2Kg and 3 Kg

(1) 12 N
(2) 24 N
(3) 36 N
(4) 48 N

5) A wedge of mass M fitted with a spring of stiffness ‘k’ is kept on a smooth horizontal surface. A rod of mass m is kept on the wedge as shown in the figure. System is in equilibrium. Assuming that

all surfaces are smooth, the potential energy stored in the spring is : 

(1) $\dfrac{mg^2\tan^2\theta}{2K}$

(2) $\dfrac{m^2g\tan^2\theta}{2K}$

(3) $\dfrac{m^2g^2\tan^2\theta}{2K}$

(4) None of these

6) A small mass slides down an inclined plane of inclination θ with the horizontal. The co-efficient of friction is $\mu = \mu_0 x$ where x is the distance through which the mass slides down and μ_0 is a constant. Then the distance covered by the mass before it stops is:

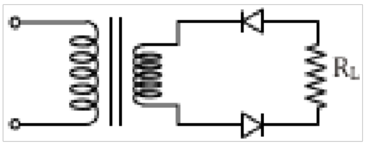
(1) $\dfrac{2}{\mu_0}\tan\theta$

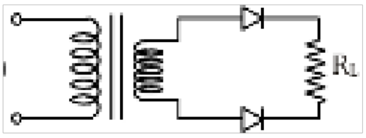
(2) $\dfrac{4}{\mu_0}\tan\theta$

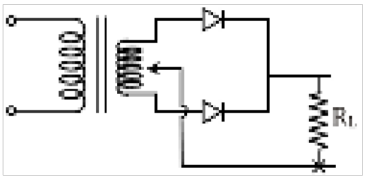
(3) $\dfrac{1}{2\mu_0}\tan\theta$

(4) $\dfrac{1}{\mu_0}\tan\theta$

7) Which of the following is not a rectifier circuit?

(1) 

(2) 

(3) 

(4) None of these

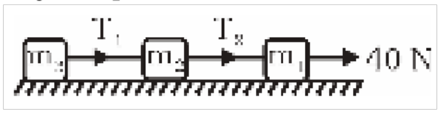
8) An unknown nucleus contains 70 neutrons and has twice the volume of the nickel $^{60}_{28}Ni$ nucleus. Identify the unknown nucleus in the form $^{A}_{Z}X$.

(1) $^{240}_{170}X$

(2) $^{110}_{40}X$
(3) $^{126}_{56}X$
(4) $^{120}_{50}X$

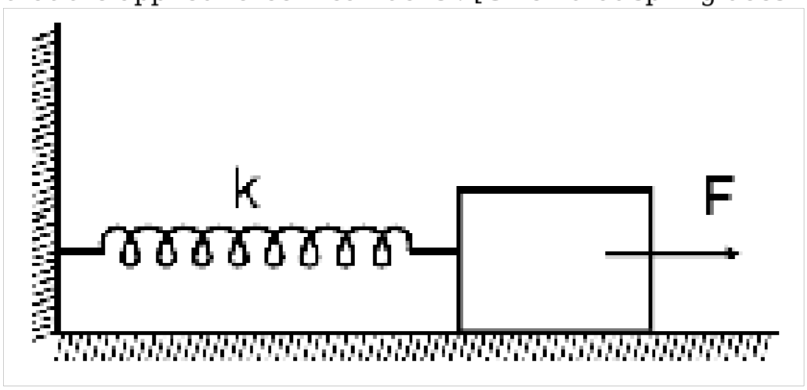
9) In an inelastic collision

(1) Only momentum is conserved
(2) Only kinetic energy is conserved
(3) Neither momentum nor kinetic energy is conserved
(4) Both momentum and kinetic energy are conserved

10) Three blocks of masses m_1 ,m_2 and m_3 are connected by massless strings as shown in the figure on a frictionless table. They are pulled with a force of 40 N. If m_1 = 10 kg, m_2 = 6 kg and m_3 = 4 kg, then tension T_2 will be : 

(1) 10 N
(2) 20 N
(3) 32 N
(4) 40 N

11) A block attached to a spring, pulled by a constant horizontal force, is kept on a smooth surface as shown in the figure. Initially, the spring is in the natural state. Then the maximum positive work that the applied force F can do is : [Given that spring does not break]



(1) $\frac{F^2}{K}$
(2) $\frac{2F^2}{K}$
(3) ∞
(4) $\frac{F^2}{2K}$

12) **Assertion (A) :** Work done by static friction may be positive.
Reason (R) : Static friction may act along direction of motion of an object.

(1) Both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**

(2) **Assertion** is correct but **Reason** is incorrect.

(3) Both **Assertion** and **Reason** are incorrect

(4) Both **Assertion** and **Reason** are correct but **Reason** is NOT the correct explanation of **Assertion**.

13) The potential energy of a particle in a force field is U = $\frac{A}{r^3} - \frac{B}{r^2}$ where A and B are positive constants and r is the distance of particle from the centre of field. For the equilibrium the distance of particle is :-

(1) $\frac{2A}{3B}$

(2) $\frac{3A}{2B}$

(3) $\frac{A}{3B}$

(4) $\frac{A}{2B}$

14) The resistivity of a pure semiconductor is 0.5 Ωm. If the electron and hole mobility be 0.39 m^2/V-s and 0.19 m^2/V-s respectively then calculate the intrinsic carrier concentration.

(1) $2.16 \times 10^{19}/m^3$

(2) $4.32 \times 10^{19}/m^3$

(3) $10^{20}/m^3$

(4) None of these

15) Which of the following statement (s) are correct ?
(a) If there is a displacement of point of application of force for a non-zero force, then work of that force must be non-zero.
(b) Work done by Normal contact force is always zero.
(c) Conservation of mechanical Energy is a special case of Work-Energy theorem. (d) Change in potential Energy is independent of frame of reference.

(1) b, d

(2) a, b, c, d

(3) c, d

(4) b, c, d

CHEMISTRY

SECTION-1

1) Which of the following has highest melting point?

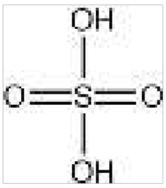
(1) CaF_2

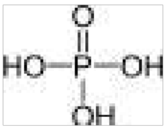
(2) $CaCl_2$

(3) $CaBr_2$

(4) CaI_2

2) Identify the incorrect structure

(1) 

(2) 

(3) 

(4) 

3) Which of the following is hypervalent molecule:

(1) BF_3

(2) IF_7

(3) CCl_4

(4) H_2O

4) Which is the CORRECT thermal stability order for H_2E (E = O, S, Se, Te and Po)?

(1) $H_2S < H_2O < H_2Se < H_2Te < H_2Po$

(2) $H_2O < H_2S < H_2Se < H_2Te < H_2Po$

(3) $H_2Po < H_2Te < H_2Se < H_2S < H_2O$

(4) $H_2Se < H_2Te < H_2Po < H_2O < H_2S$

5) Bond order of O_2^{2-} is :-

(1) 2

(2) 1

(3) 3

(4) 1.5

6) The hybrid orbitals used by P in PCl_5 are :-

(1) s, p_x, p_y, d_{z^2}, d_{xy}

(2) s, p_x, p_y, d_{z^2}, f

(3) s, p_x p_y, p_z, d_{z^2}

(4) None of these

7)

The correct order of the O–O bond length in O_2, H_2O_2 and O_3 is :-

(1) $O_3 > H_2O_2 > O_2$
(2) $O_2 > H_2O_2 > O_3$
(3) $O_2 > O_3 > H_2O_2$
(4) $H_2O_2 > O_3 > O_2$

8) Give the correct order of strength of H-bonding
(a) O–H ----- O (b) F–H ------ F
(c) Cl–H ----- Cl (d) N–H ----- N

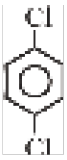
(1) a > b > c > d
(2) b > a > c > d
(3) b > a > d > c
(4) b > a > c = d

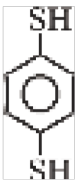
9) How many π-bonds are present in C_2 molecule?

(1) 1
(2) 2
(3) 0
(4) 3

10) Which of the following has zero dipole moment

(1) 

(2) 

(3) 

(4) All of the above

11) Which of the following pairs of ions are isoelectronic and isostructural ?

(1) ClO_3^-, CO_3^{2-}
(2) SO_3^{2-}, NO_3^-

(3) ClO_3^-, SO_3^{2-}

(4) CO_3^{2-}, SO_3^{2-}

12) Which species would have maximum hydration ?

(1) Be^{+2}

(2) Li^+

(3) Mg^{+2}

(4) NH_4^+

13) Which is not correctly matched.

	Column-I	Column-II
(1)	Vitamin A	Night Blindness
(2)	Vitamin B_6	Convulsions
(3)	Vitamin C	Scurvy
(4)	Vitamin D	Muscular weakness

(1) 1

(2) 2

(3) 3

(4) 4

14) At the isoelectric point amino acids are present as :

(1) $H_2NCHRCOOH_2^+$

(2) $H_3N^+CHRCOO^-$

(3) $H_3N^+CHRCOOH$

(4) $H_2NCHRCOO^-$

15) Glycogen is a branched chain polymer of
α-D-glucose units in which chain is formed by C_1–C_4 glycosidic linkage whereas branching occurs by the formation of C_1–C_6 glycosidic linkage. Structure of glycogen is similar to.

(1) Fructose

(2) Amylopectin

(3) Cellulose

(4) Glucose

16) Which of the following statement(s) is/are correct
(i) Glucose is reducing sugar
(ii) Sucrose is reducing sugar
(iii) Maltose is non reducing sugar
(iv) Lactose is reducing sugar

(1) i and ii only
(2) i and iii only
(3) i and iv only
(4) All of these

17) Which of the statements about 'Denaturation' given below are correct?
(i) Denaturation of proteins causes loss of secondary and tertiary structure of the protein.
(ii) Denaturation leads of the conversion of double strand of DNA into single strand
(iii) In denaturation, primary structure remains intact

(1) ii and iii only
(2) i and iii only
(3) i and ii only
(4) i, ii and iii

18) X $\xleftarrow[\Delta]{HI}$ Glucose $\xrightarrow{HNO_3}$ Y What are X and Y respectively?

(1) n-hexane, Gluconic acid
(2) Gluconic acid, Saccharic acid
(3) Gluconic acid, n–hexane
(4) n–hexane, Saccharic acid

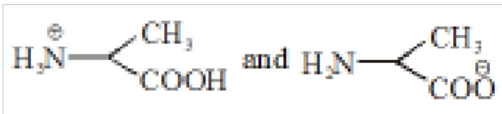
19) Lactose is made of :

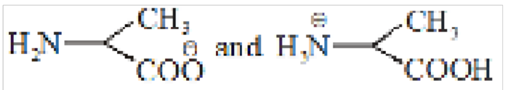
(1) α-D-glucose only
(2) α-D-glucose and β-D-glucose
(3) β-D-galactose and β-D-glucose
(4) α-D-galactose and β-D-glucose

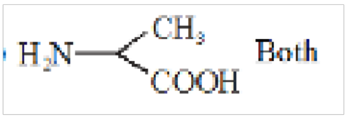
20) Molecule following Lewis octet rule is :

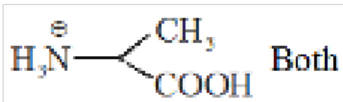
(1) BCl_3
(2) PCl_3
(3) $AlCl_3$
(4) ICl_3

21) Alanine exist as zwitter ion at pH = 6.02. Structure of alanine at pH = 2 and pH = 10 are respectively :-

(1) 

(2) 

(3) 

(4) 

22) Anomers of glucose differ in configuration at :-

(1) C_1
(2) C_2
(3) C_3
(4) C_4

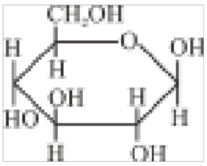
23) Select the incorrect statement:-

(1) Lactose is commonly known as milk sugar.
(2) Glycogen is commonly known as Animal starch
(3) Starch consist of two components-amylose & amylopectin
(4) Cell wall of bacteria is made up of glycogen.

24) Which among the following is non essential amino acid ?

(1) Methionine
(2) Isoleucine
(3) Histidine
(4) Proline

25) Which is correct statement ?

(1) Maltose is composed of two β-D-Glucose units combined with 1–4 glycosidic link.
(2) 
is a structure of α-D-(+) Glucopyranose
(3) Glycogen is similar to amylopectin's structure which is branched chain polymer of β-D-Glucose
(4) Cellulose is straight chain polymer of β-D-Glucose.

26) Which of following has maximum number of $d\pi$-$p\pi$ bonding :

(1) CO_2
(2) SO_4^{2-}
(3) ClO_4^-
(4) SO_2

27) Which of following represent incorrect order of bond strength :-

(1) Cl–Cl > Br–Br > F–F > I–I
(2) CO > NO
(3) $ClO^- < ClO_2^- < ClO_3^-$
(4) Si–Si > C–C > Ge–Ge

28) Which is not correctly matched ?

(1) XeO_3 – Trigonal bipyramidal
(2) ClF_3 – T shape
(3) XeF_2 – Linear
(4) IF_7 – Pentagonal bipyramidal

29) Correct order of bond angle is:-

(1) $BF_3 < BCl_3 < BBr_3$
(2) $SO_2 > SO_3$
(3) $NO_2 < NO_2^-$
(4) $NH_3 > PH_3$

30) The fluorine molecule is formed by :

(1) p-p orbitals (sideways overlap)
(2) p-p orbitals (end-to-end overlap)
(3) sp-sp orbitals
(4) s-s orbitals

31) Identify the hybridisation of central atom in I_3^- ion :-

(1) sp^3
(2) sp^3d
(3) sp^3d^2
(4) sp

32) Among the following, the molecule that is linear is :

(1) CO_2
(2) NO_2
(3) SO_2
(4) ClO_2

33) Bond order of O_3 are respectively :-

(1) 2

(2) 1.5
(3) 1
(4) 3

34) Shape of 'XeO_2F_2' molecule is :-

(1) Tetrahedral
(2) See Saw
(3) Square planar
(4) Distorted octahedral

35) The molecule which is having maximum no. of lone pair at central atom:-

(1) SiF_4
(2) BrF_3
(3) XeF_2
(4) SF_4

SECTION-2

1) Which molecule have central atom with 20%
s-character hybridised orbital

(1) CH_4
(2) PCl_5
(3) BCl_3
(4) XeO_3

2)

Shape of XeF_4 molecule is :-

(1) Linear
(2) Pyramidal
(3) Tetrahedral
(4) Square planar

3) Which of the following combination of orbitals does not form any type of covalent bond?

(1) $p_z + p_z$
(2) $p_y + p_y$
(3) $p_x + p_y$
(4) $s + s$

4) Shape of $[H_3O]^+$ is :-

(1) Tetrahedral
(2) Angular
(3) Pyramidal
(4) Trigonal planar

5) A molecule with 2 bond pair and 2 lone pair on central atom is likely to be :-

(1) V shaped
(2) Pyramidal
(3) Square planar
(4) Triagonal pyramidal

6) Hybridization of the central atom will change when :-

(1) NH_3 combines with H^+
(2) H_3BO_3 combines with OH^-
(3) NH_3 forms NH_2^-
(4) H_2O combines with H^+

7) Choose the correct order of C—C bond length in the given compounds :

(1) Ethylene < Acetylene < Benzene < Ethane
(2) Acetylene < Ethylene < Benzene < Ethane
(3) Acetylene < Benzene < Ethylene < Ethane
(4) Acetylene < Benzene < Ethane < Ethylene

8) Match the following

	Column-I (Species)		Column-II (Bond order)
a	H_2^+	p	0
b	He_2	q	1.5
c	N_2^+	r	0.5
d	O_2^-	s	2.5

(1) a-r, b-p , c-s, d-q
(2) a-p, b-r , c-s, d-q
(3) a-r, b-p , c-q, d-s
(4) a-p, b-r , c-q, d-s

9) Which is having highest bond angle -

(1) PCl_3
(2) PBr_3

(3) PF_3

(4) PI_3

10) Which of the following relation represent a stable molecule in respect of MOT ?

(1) $N_b > N_a$

(2) $N_b < N_a$

(3) $N_b = N_a$

(4) None of these

11) Compare the dipole moment of following molecules-

(1) $H_2O < H_2S < CO_2$

(2) HF < HCl < HBr

(3) $NH_3 > NF_3 > BF_3$

(4) $CH_4 > CHCl_3 > CCl_4$

12) Ionic bond forms more easily between elements with comparatively.

(1) Low I.E. & high value of electron affinity

(2) High I.E. & high negative value of EGE

(3) High I.E. & low negative value of EGE

(4) Low I.E. & low value of electron affinity

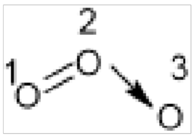
13) **Assertion :-** PF_5, SF_6, H_2SO_4 are considered as hypervalent molecules
Reason :- There are 10, 12, 12 electrons around P, S & S in PF_5, SF_6 and H_2SO_4 molecules respectively.

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

14) Find correct set of formal charge on marked oxygen respectively. 

(1) +1, +1, +1

(2) +1, –1, 0

(3) +1, 0, –1

(4) 0, +1, –1

15) Which of following will not exist :-

(1) NF_5

(2) BiI_5

(3) PbI_4

(4) All

BOTANY

SECTION-1

1) Mark the **incorrect** statement

(1) Isolated metabolic reactions in vitro are not living things but surely living reactions
(2) In yeast and hydra reproduction occurs by budding.
(3) No non-living object exhibits growth
(4) Cellular organisation of the body is the defining feature of life forms

2) In *Mangifera indica* Linn, the specific epithet is

(1) *Mangifera*
(2) *indica*
(3) Linn.
(4) Both 1 and 2

3) Growth and reproduction are mutually exclusive events in

(1) Bacteria and protista
(2) Protista and unicellular fungi
(3) Yeast and paramecium
(4) Higher plants and animals

4) Which of the following set of organisms reproduces by fragmentation (asexual mode of reproduction)?

(1) *Amoeba*, fungi and earthworm
(2) Fungi, filamentous algae and protonema of mosses
(3) *Hydra*, fungi, *Amoeba* and bacteria
(4) Earthworm, bacteria and fungi

5) Photoperiod affects reproduction in :-

(1) Animals
(2) Plants
(3) Seasonal breeders
(4) All of these

6) All living organisms can be classified into different taxa. This process of classification is:-

(1) Systematics
(2) Taxonomy
(3) Nomenclature
(4) Identification

7) Chemosynthetic autotrophic bacteria play a great role in recycling nutrients like:-

(1) Only Nitrogen
(2) Nitrogen, phosphorous and calcium
(3) Nitrogen, phosphorous, iron and sulphur
(4) Only sulphur

8) Which option is related with only and only monera and not related with other kingdom of R. H. Whittaker ?

(1) Decomposer nature
(2) Presence of cell wall and cell membrene
(3) Well develop tissue system
(4) Nitrogen fixation ability with prokaryotic nature

9)

Select the **wrong** statement.

(1) The walls of diatoms are easily destructible.
(2) Diatomaceous earth is formed by the cell walls of diatoms.
(3) Diatoms are chief producers in the oceans.
(4) Diatoms are microscopic and float passively in water.

10) Aristotle made the classification of plants by using ?

(1) Internal morphological characters
(2) Simple morphological characters
(3) Histological characters
(4) Phylogenetic characters

11) How many organisms are included in Monera in given list ?
(A) Slime moulds (B) Dinoflagellates
(C) Protozoans (D) Basidiomycetes
(E) Archebacteria (F) Blue-green algae

(1) 2
(2) 4
(3) 3
(4) 5

12) Which of the following are responsible for the production of methane in the gut of several

ruminant animals ?

(1) Halophiles
(2) Thermoacidophiles
(3) *Thermus aquaticus*
(4) Methanogens

13) A bacterium divides in every 20 min. If a culture containing 10^5 cells/ml is grown for 2 hours then what will be the number of cells in the culture.

(1) 32×10^5 cells/ml
(2) 64×10^5 cells/ml
(3) 16×10^5 cells/ml
(4) 20×10^5 cells/ml

14) Specialized cell of nostoc and anabaena fix nitrogen are known as

(1) Cyst
(2) Heterocyst
(3) Oocytes
(4) Cholecystitis

15) Select the correct option w.r.t. animal disease caused by bacteria:

(1) Cholera
(2) Typhoid
(3) Citrus canker
(4) Both 1 and 2

16) Find out **incorrect** with respect to kingdom protista :

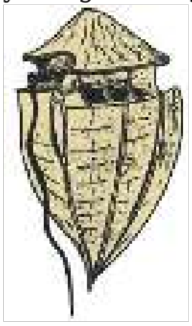
(1) Diatoms have silica cell wall
(2) Dinoflagellates have two flagella
(3) Englena possess well developed cell wall
(4) Slime mould are without chlorophyll

17) Two unequal flagella are found in :

(1) *Amoeba*
(2) Diatoms
(3) Euglenoids
(4) Slime mould

18) Identify the given figure and select the **correct** option. 

(1) It is marine phyto plankton
(2) It is a saprophytic protists
(3) It is parasitic predator believed to be primary relative of animals
(4) Ciliated protozoan

19) Identify the given organism in the following diagram and select the important character of this organism. 

	Organism	Character
(1)	Diatom	Presence of two over lapping shells
(2)	Dinoflagellate	Presence of stiff cellulose plates
(3)	Euglenoid	Presence of pellicle
(4)	Slime mould	Presence of true walls in spores

(1) 1
(2) 2
(3) 3
(4) 4

20) Which fungi causes rust and smut diseases respectively?

(1) *Ustilago* and *Agaricus*
(2) *Puccinia* and *Ustilago*
(3) *Puccinia* and *Agaricus*
(4) *Ustilago* and *Puccinia*

21) Which one of the following is mismatched :-

(1) *Penicillium* - Source of antibiotics

(2) *Albugo* - Parasitic fungi on legume

(3) *Neurospora* - Used in biochemical and genetic work

(4) *Agaricus* - Edible fungus

22) Dikaryon means :-

(1) Two chromosomes per cell
(2) Two mitochondria per cell
(3) Two nuclei per cell
(4) Two spores formed by a cell

23) The suppression of fusion of gametes is a general trend among fungi. However, sexual reproduction has **not** been noticed at all in :-

(1) Ascomycetes
(2) Basidiomycetes
(3) Deuteromycetes
(4) All of these

24) *Contagium vivium fluidum* was proposed by–

(1) D.J. Ivanowsky
(2) M.W. Beijerinck
(3) Stanley
(4) Robert Hooke

25) The function of phycobiont in lichens is to :

(1) Absorb water from soil
(2) Prepare food for mycobiont
(3) Absorb nutrients from dead remains
(4) Anchor the lichen body to the substratum

26) Causal agent of potato spindle tuber disease :

(1) Is abnormally folded protein
(2) Contain DNA as genetic material
(3) Were discovered by T.O. Diener
(4) Also cause mad cow disease

27) Rate of breakdown of complex organic matter will be fast in which of the following ?

(1) Warm and moist environment
(2) Anaerobic conditions
(3) Area of high temperature and less humidity

(4) Area of low temperature and high humidity

28) Standing crop is :-
(i) Mass of living organisms.
(ii) The number in a unit area.
(iii) Amount of nutrient such as nitrogen calcium in unit area.
(iv) Amount of detritus in unit area.

(1) only i
(2) i & ii
(3) i, ii, iv
(4) i, ii, iv

29) Which one of the following is **not** a function unit of an ecosystem ?

(1) Energy flow
(2) Decomposition
(3) Productivity
(4) Stratification

30) Identify the **correct** set of statements with regard to properties of humus.
(a) Highly resistant to microbial action.
(b) Dark coloured amorphous substance.
(c) End product of detritus food chain.
(d) Reservoir of nutrients.
(e) Undergoes decomposition very fast.

(1) (a), (b) and (e) only
(2) (a) and (b) only
(3) (a), (b) and (c) only
(4) (a), (b) and (d) only

31) Productivity is expressed in terms of

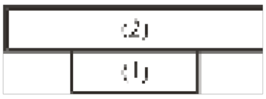
(1) $gm^{-2}yr^{-1}$
(2) $(Kcal\ m^{-2})\ yr^{-1}$
(3) $Kgm^{-2}\ yr^{-1}$
(4) Both 1 and 2

32) Plants capture only .... of the PAR and this small amount of energy sustains the entire living world.

(1) 5-10 per cent
(2) 2-10 per cent
(3) 1-5 per cent
(4) 70-80 per cent

33) Based on the source of their nutrition, organism occupy a specific place in the food chain that is know as their ______.

(1) Trophic level
(2) Standing crop
(3) Niche
(4) Standing state

34) 

Above pyramid reperesent.

(1) Pyramid of grassland on the basis of biomass.
(2) Pyramid of grassland on the basis of number.
(3) Pyramid of aquatic ecosystem on basis of biomass
(4) Pyramid of energy

35) There are certain limitations of ecological pyramids **correct** option :-
(A) It assumes a simple food chain.
(B) Saprophytes are not given any place in ecological pyramids.
(C) It does not take into account the same species belonging to two or more trophic levels.

(1) A, B Only
(2) B, C Only
(3) Only B
(4) A, B, C

SECTION-2

1) **Statement-I :-** All living organism show growth & reproduction.
Statement-II :- Growth is defining features of living.

(1) Statement-I is true but statement-II is false
(2) Both statement-I and II is true
(3) Statement-I is false but statement-II is true
(4) Both statement-I and II is false

2) Which of the following is **incorrect** ?

(1) The category 'order' is the assemblage of families which exhibit a few similar characters
(2) Classification is branch of biology which deals with the principles and procedures of identification and nomenclature of organisms
(3) Systematics takes into account the evolutionary relationships between organisms also
(4) The binomial nomenclature system was given by Carolus Linnaeus

3) Match the **column-I** with **column-II**.

	Column-I		Column-II
(A)	Family	(i)	Muscidae
(B)	Order	(ii)	Sapindales
(C	Class	(iii)	Dicotyledonae
(D)	Genus	(iv)	*Triticum*

(1) A-i, B-ii, C-iv, D-iii
(2) A-iii, B-ii, C-i, D-iv
(3) A-i, B-ii, C-iii, D-iv
(4) A-iv, B-ii, C-iii, D-i

4) Which of the following set of taxa cover greater number of organisms?

(1) Phylum and family
(2) Family and Order
(3) Class and Phylum
(4) Super Family and Class

5) Which out of the following was **not** the basis of the five-kingdom classification?

(1) Cell structure
(2) Gross morphology
(3) Mode of nutrition
(4) Phylogenetic relationships

6) Which of the following is **incorrect** about blue green algae ?

(1) They may be unicellular, colonial or filamentous
(2) Their colonies are generally surrounded with membranous sheath
(3) They often form blooms in polluted water bodies
(4) Some of the members can fix atmospheric nitrogen

7) Consider the following four statements (A-D) and select the option which includes all the **correct** ones only.
(A) Archaebacteria differ from other bacteria in having a different cell wall structure
(B) Bacteria as a group show the most extensive metabolic diversity
(C) Bacterial structure is very complex, they are very simple in behavior.
(D) Mycoplasma can survive without oxygen.

(1) Statements B, C & D
(2) Statements A, C only
(3) Statements B, D only
(4) Statements A, B & D

8) Dinoflagellates :

(1) Are mostly marine and photosynthetic

(2) Have cell wall made up of two cellulosic shells

(3) Are usually have two cilia, one transverse and one longitudinal

(4) 1 and 3 both

9) Select the mismatched pair :

(1) *Flagellated protozoan - Trypanosoma*

(2) *Sporozoans - Plasmodium*

(3) *Amoeboid protozoans - Gonyaulex*

(4) *Ciliated protozoans - Paramoecium*

10) Read the following point :-
(A) Body septate and branched
(B) Some members are saprophytes or parasites
(C) Help in mineral cycling
(D) Reproduce by asexual spores known as conidia
Above points are related with which class of organisms

(1) Phycomycetes

(2) Euglenoids

(3) Deuteromycetes

(4) Basidiomycetes

11) **Statement I :** Lichens are very good pollution indicators.
Statement II : Lichens do not grow in polluted areas.

(1) Both statement-I and statement-II are incorrect

(2) Both statement-I and statement-II are correct

(3) Statement-I is incorrect and statement-II is correct

(4) Statement-I is correct and statement-II is incorrect

12) Which of the following is **not** a favourable factor for increasing the rate of decomposition ?

(1) Oxygen richness of soil.

(2) Detritus rich in lignin

(3) Low temperature

(4) High soil moisture

13) Which of the following statements are **correct** ?
(a) Net primary productivity is available biomass for the consumption to heterotrophs
(b) Secondary productivity is defined as rate of formation of new organic matter by consumers
(c) The annual net primary productivity of ocean is greater than land
(d) The rate of biomass production is called productivity

(1) a, b, c & d

(2) a, b & c

(3) a, b & d

(4) a & c

14) Choose the number of **correct** features associated with energy flow in an ecosystem :-
A. Grazing food chain is major conduit for energy flow in terrestrial ecosystem.
B. Pyramid of energy is always upright.
C. Ecosystems are not exempted from the IInd law of thermodynamics
D. Detritus food chain made up of decomposers which are heterotrophic organisms, mainly fungi and bacteria.

(1) 3

(2) 1

(3) 2

(4) 4

15) **Statement 1:** Net primary productivity is less than the gross primary productivity.
Statement 2: Net primary productivity is equal to the gross primary productivity minus the respiration losses.

(1) Both statements 1 and 2 are correct.

(2) Both statements 1 and 2 are incorrect

(3) Statement 1 is correct and statement 2 is incorrect.

(4) Statements 1 is incorrect and statements 2 is correct.

ZOOLOGY

SECTION-1

1) Which step proved to be the main challenging obstacle in the production of human insulin by genetic engineering ?

(1) Splitting A & B polypeptide chains.

(2) Addition of C-peptides to pro-insulin.

(3) Getting insulin assembled into mature form.

(4) Removal of C-peptide from active insulin.

2) What is the main reason for developing laws by some nations in the context of biodiversity and traditional knowledge?

(1) To increase their financial wealth

(2) To prevent unauthorised exploitation of their bio-resources

(3) To enhance their industrialisation

(4) To reduce the biodiversity richness

3) The treatment of SCID can be done by:

(1) Through RNA interference

(2) Through introduction of engineered lymphocytes with functional ADA cDNA effectively by using retroviral vector

(3) By introduction of ADA enzyme in early embryonic stage

(4) By preventing translation of ADA gene

4) **Statement I:** Basmati rice has over 27 documented varieties are grown in India.
Statement II: In 1997, an American company obtained patent rights on Basmati rice, which allowed them to sell a 'new' variety derived from Indian farmer’s varieties both in the US and abroad.

(1) Statement I is false but Statement II is true

(2) Both Statement I and Statement II are true.

(3) Both Statement I and Statement II are false

(4) Statement I is true but Statement II is false

5) Given below are two statements:
Statement I: Using Agrobacterium vectors, nematode-specific genes were introduced into the host plant to produce both sense and anti-sense RNA, which formed a double stranded (dsRNA) that initiated RNA interference (RNAi) and silenced the specific mRNA of the nematode.
Statement II: The transgenic plant expressing specific interfering RNA could not protect itself from the parasite nematode.
In the light of the above statements, choose the correct answer from the options given below.

(1) Statement I is false but Statement II is true

(2) Both Statement I and Statement II are true

(3) Both Statement I and Statement II are false

(4) Statement I is true but Statement II is false

6) Match the columns -

	Column-I		**Column-II**
(A)	Rosie	(i)	α-1 antitrypsin
(B)	ELISA	(ii)	Protein enriched milk
(C)	rop	(iii)	Test to detect antigen or antibody
(D)	Emphysema	(iv)	Codes for protein involved in plasmid replication

(1) A-ii, B-iii, C-iv, D-i

(2) A-i, B-iii, C-iv, D-ii

(3) A-ii, B-iii, C-i, D-iv

(4) A-iv, B-iii, C-ii, D-i

7) Protein encoded by a gene cryI Ab control :-

(1) Corn borer

(2) Cotton boll worm

(3) Beetles

(4) Mosquitoes

8) If you create a GM crop, which agency of Government of India will you approach to get permission for its commercial use?

(1) IARI
(2) CDRI
(3) GEAC
(4) KVIC

9) Erythroblastosis foetalis is a disease in which :-

(1) Adult have severe anaemia and jaundice
(2) Female have severe anaemia and jaundice
(3) Male have severe anaemia and jaundice
(4) Foetus have severe anaemia and jaundice

10) Lymph capillaries of intestine villi are called as:-

(1) Lymphatics
(2) Portal vein
(3) Thoracic duct
(4) Lacteals

11) Match the columns :-

Column-I		**Column-II**	
(a)	Heart failure	(i)	Pumping activity of heart decreases
(b)	Heart attack	(ii)	Heart muscles damaged due to inadequate blood supply
(c)	Cardiac arrest	(iii)	Heart stops beating

(1) a-i, b-ii, c-iii
(2) a-i, b-iii, c-ii
(3) a-iii, b-ii, c-i
(4) a-ii, b-i, c-iii

12) Which of the following correctly explains a phase/event in cardiac cycle in a standard electrocardiogram ?

(1) QRS complex indicates atrial contraction
(2) QRS complex indicates ventricular contraction
(3) Time between S and T represents atrial systole
(4) P-wave indicates the beginning of ventricular contraction

13) Which one is the correct route through which impulse travels in the heart ?

(1) SA node → Purkinje fibres → Bundle of His → AV node → Heart muscles

(2) AV node → SA node → Purkinje fibres → Bundle of His → Heart muscles
(3) AV node → Bundle of His → SA node → Purkinje fibres → Heart muscles
(4) SA node → AV node → Bundle of His → Purkinje fibres → Heart muscles

14) "Bundle of His" is a network of :-

(1) Muscle fibres distributed throughout the heart walls
(2) Muscle fibres found only in the ventricle wall
(3) Nerve fibres distributed in ventricles
(4) Nerve fibres found throughout the heart

15) In a portal system of man :-

(1) A vein starts from an organ and ends in heart
(2) A vein enters into organ other than heart and breaks in capillaries
(3) An artery breaks in an organ and restarts by union of its capillaries
(4) Blood from intestine is brought into the kidneys and then in Inferior vena cava

16) Select the answer with incorrect match :-

Column-I		Column-II	Column-III
(1)	Fishes	2-chambered heart	An atrium and a venticle
(2)	Amphibians	3-chambered heart	An atrium and two ventricles
(3)	Crocodiles	4-chambered heart	Two atria and two ventricles
(4)	Aves	4-chambered heart	Two atria and two ventricles

(1) 1
(2) 2
(3) 3
(4) 4

17) Set of phagocytic cells among the following is

(1) Neurophils and platelets
(2) Neurophils and platelets
(3) Monocytes and neutrophils
(4) Lymphocytes and monocytes

18) Lymph is modified blood that contains :-

(1) RBC and WBC

(2) RBC, WBC and proteins
(3) WBC and all proteins
(4) All contents of blood except RBC, platelets and certain proteins

19) Correct match the following columns :-

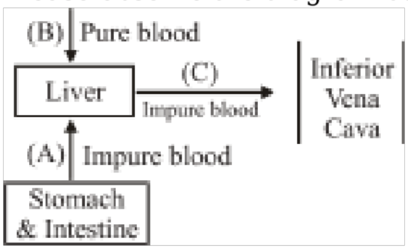
Column-I		Column-II	
(A)	Neutrophils	I.	20-25% of WBC
(B)	Basophils	II.	2-3% of WBC
(C)	Monocytes	III.	6-8% of WBC
(D)	Eosinophils	IV.	0.5-1% of WBC
(E)	Lymphocytes	V.	60-65% of WBC

(1) A-I, B-III, C-IV, D-II, E-V
(2) A-V, B-IV, C-III, D-II, E-I
(3) A-V, B-IV, C-I, D-II, E-III
(4) A-V, B-II, C-IV, D-III, E-I

20) Which structures are directly involved in the pulmonary circulation ?

(1) Right atrium, aorta and left ventricle
(2) Left ventricle, aorta and inferior vena cava
(3) Superior vena cava, right atrium and left atrium
(4) Right ventricle, pulmonary arteries and left atrium

21)

Please observe the diagram below


Here vessel A, B and C are –

(1) A – Hepatic vein
B – Hepatic portal vein C – Hepatic artery
(2) A – Hepatic portal vein
B – Hepatic artery C – Hepatic vein
(3) A – Hepatic artery B – Hepatic vein
C – Hepatic portal vein
(4) A – Hepatic portal vein B – Hepatic vein C – Hepatic artery

22) Blood cell called cell fragment is :-

(1) RBCs
(2) WBCs
(3) Spindle cell
(4) Platelets

23) Coronary artery disease affects the vessels that supply blood to :-

(1) Kidney
(2) Liver
(3) Lungs
(4) Heart muscles

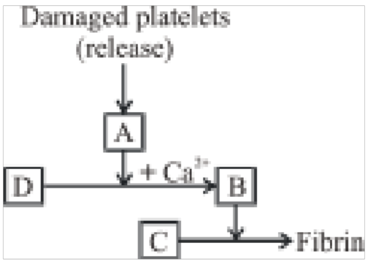
24) Assertion : Blood group O^{-ve} is universal donor
Reason : Blood group O^{-ve} contain both antigen A and antigen B.

(1) Both Assertion and reason are true.
(2) Both Assertion and reason are false.
(3) Assertion is true but reason is false.
(4) Assertion is false but reason is true.

25) Mammalian RBC are :-

(1) Biconcave, circular , non nucleated
(2) Biconvex, Nucleated
(3) Oval, Nucleated
(4) Biconcave and nucleated

26) Identify A, B and C in the given below blood clotting process.



	A	B	C
(1)	Thromboplastin	Prothrombin	Fibrinogen
(2)	Thrombin	Fibrinogen	Thrombo-Kinase
(3)	Thromboplastin	Thrombin	Fibrinogen
(4)	Prothrombin	Thrombin	Fibrinogen

(1) 1
(2) 2
(3) 3
(4) 4

27) Which of the following conditions cause erythroblastosis foetalis ?

(1) Mother Rh^{+ve} and foetus Rh^{-ve}
(2) Mother Rh^{-ve} and foetus Rh^{+ve}
(3) Both mother and foetus Rh^{-ve}
(4) Both mother and foetus Rh^{+ve}

28) Volume of air that will remain in lungs after a forceful expiration is :-

(1) RV
(2) FRC
(3) EC
(4) IC

29) What is the partial pressure of carbon dioxide and oxygen in atmospheric air ?

(1) pCO_2 159 mm Hg, pO_2 40 mm Hg
(2) pCO_2 0.3 mm Hg, pO_2 159 mm Hg
(3) pCO_2 40 mm Hg, pO_2 45 mm Hg
(4) pCO_2 45 mm Hg, pO_2 116 mm Hg

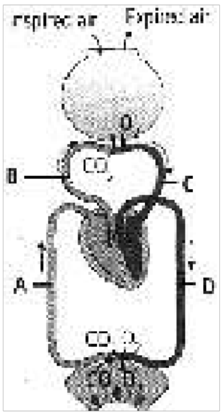
30) The enzyme carbonic anhydrase is found in :-

(1) Tissue cells and plasma
(2) Tissue cells and erythrocytes
(3) Erythrocytes
(4) WBC

31) During normal expiration :-

(1) EICM and diaphragm muscles contract
(2) IICM and abdominal muscles contract
(3) EICM and muscles of diaphragm muscles relax
(4) Intrapulmonary pressure is less than atmospheric pressure

32) The given figure shows diagrammatic representation of exchange of gases at the alveolus and the body tissues with blood and transport of oxygen and carbon dioxide.



Identify the blood vessels A to D.

	A	B	C	D
(1)	Systemic vein	Pulmonary artery	Pulmonary vein	Systemic artery
(2)	Systemic artery	Pulmonary artery	Pulmonary vein	Systemic vein
(3)	Pulmonary artery	Systemic vein	Pulmonary vein	Systemic artery
(4)	Systemic vein	Pulmonary vein	Pulmonary artery	Systemic artery

(1) 1
(2) 2
(3) 3
(4) 4

33)

About 97% of oxygen is transported by RBC. The remaining 3% is :-

(1) Retained in lungs
(2) Dissolved in plasma and transported
(3) Attached to cell membrane
(4) Inside mitochondria

34) Functional residual capacity in human is the amount of air _______.

(1) That can be breathout from lungs after forceful expiration.
(2) That remains in lungs after normal expiration.
(3) That can be filled in lungs by forceful inspiration.
(4) That remains in lungs after forceful expiration.

35) Arrange the following in the order of decreasing volume :-
(A) Tidal volume
(B) Residual volume
(C) Inspiratory reserve volume

(D) Vital capacity

(1) D → C → B → A
(2) D → B → C → A
(3) B → C → D → A
(4) C → B → D → A

SECTION-2

1) Carefully read the following and name the event
Diaphragm → flat, ElCM - contraction, Ribs → outwards :-

(1) Expiration
(2) Inspiration
(3) Both
(4) None

2) Some symptoms of a disorder are given read them and identify the disorder :-
(a) Alveolar walls are damaged
(b) Respiratory surface is decreased
(c) Smoking is major cause

(1) Asthma
(2) Emphysema
(3) Silicosis
(4) Pneumonia

3) Primary sites of exchange of gases are :-

(1) Alveoli
(2) Tissue
(3) Terminal bronchiole
(4) Initial bronchiole

4)

Which enzyme is present in RBC to catalyse the given reactions ?
$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$

(1) Catalase
(2) Peroxidase
(3) Carboxypeptidase
(4) Carbonic anhydrase

5) Chemosensitive area is present in :-

(1) Pons
(2) Medulla
(3) Pneumotaxic centre
(4) Cerebrum

6) In the given statements choose the odd one :-

(1) Thoracic chamber is anatomically an air-tight chamber
(2) Any change in volume of the thoracic cavity is reflected in the lung (pulmonary) cavity
(3) The inner pleural membrane is in close contact with thoracic lining
(4) The inner pleural membrane is in cotact with the lung surface

7)

Asthma is caused due to:-

(1) Infection of lungs
(2) Infection of trachea
(3) Spasm in bronchial muscles
(4) Bleeding into pleural cavity

8) Which of the following option gives correct information about the coverings of thoracic cage on various surfaces :-

	Anterior Surface	**Posterior Surface**	**Dorsal Surface**
(1)	Sternum and Ribs	Ribs	Diaphragm
(2)	Sternum and Ribs	Diaphragm	Clavicle Bones and Neck
(3)	Sternum and Ribs	Vertebral Column	Clavicle Bones and Neck
(4)	Clavicle Bones and Neck	Diaphragm	Vertebral Column

(1) 1
(2) 2
(3) 3
(4) 4

9) Gaseous exchange at alveoli depends upon ______

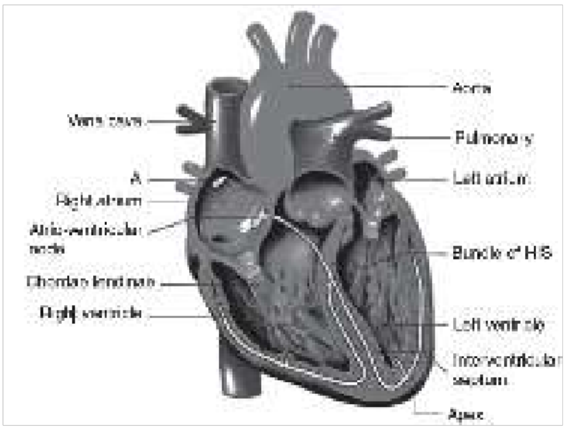
(1) Solubility of gases
(2) Partial pressure of gases
(3) Thickness of respiratory surfaces
(4) All of these

10) Select the incorrect match

(1) Inspiratory capacity – (TLC – FRC)
(2) Expiratory capacity – [(FRC – RV) + TV]
(3) Vital capacity – (IRV + EC)
(4) Total lung capacity – (VC – RV)

11) Clot is formed mainly by the network of threads called :-

(1) Fibrinogen
(2) Prothrombin
(3) Thrombin
(4) Fibrin

12) The full form of label ‘A’ in the diagram is 

(1) Sin–Atrial–Node
(2) Sinu–Atrial–Node
(3) Sino–Atrial–Node
(4) Sinus–Atrial–Node

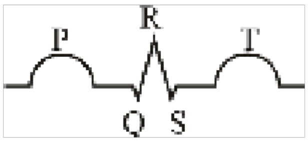
13) Apex of heart is

(1) Upper and tilted toward left
(2) Lower and tilted towards left
(3) Upper and tilted towards right
(4) Lower and tilted towards right

14) The heart sound 'Dup' is produced when :-

(1) Semi lunar valve is open
(2) Mitral valve is open
(3) Tricuspid valve is open
(4) Semilunar valve closed.

15) The diagram given here is the standard ECG of normal person. The P-wave represents :-



(1) End of systole

(2) Depolarisation of both the atria

(3) Initiation of the ventricular contraction

(4) Beginning of the systole in ventricle

ANSWER KEYS

PHYSICS

SECTION-1

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	1	3	3	3	3	4	3	4	1	4	2	3	2	1	1	1	1	3	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35					
A.	1	1	1	1	3	4	2	1	3	3	3	3	2	3	3					

SECTION-2

Q.	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
A.	3	2	1	2	3	1	2	4	1	2	2	1	2	1	3

CHEMISTRY

SECTION-1

Q.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	1	4	2	3	2	3	4	3	2	2	3	1	4	2	2	3	4	4	3	2
Q.	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85					
A.	1	1	4	4	4	3	4	1	4	2	2	1	2	2	3					

SECTION-2

Q.	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
A.	2	4	3	3	1	2	2	1	4	1	3	1	1	4	4

BOTANY

SECTION-1

Q.	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	3	2	4	2	4	2	3	4	1	2	1	4	2	2	4	3	3	2	2	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135					
A.	2	3	3	2	2	3	1	2	4	4	4	2	1	3	4					

SECTION-2

Q.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	2	3	3	2	2	4	1	3	3	2	2	3	1	1

ZOOLOGY

Telegram: @NEETxNOAH

SECTION-1

Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	**3**	**2**	**2**	**2**	**4**	**1**	**1**	**3**	**4**	**4**	**1**	**2**	**4**	**2**	**2**	**2**	**3**	**4**	**2**	**4**

Q.	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185
A.	**2**	**4**	**4**	**3**	**1**	**3**	**2**	**1**	**2**	**3**	**3**	**1**	**2**	**2**	**1**

SECTION-2

Q.	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200
A.	**2**	**2**	**1**	**4**	**2**	**3**	**3**	**4**	**4**	**4**	**4**	**3**	**2**	**4**	**2**

SOLUTIONS

PHYSICS

1) $\vec{0} = m\vec{v}_1 + m\vec{v}_2 \; \vec{v}_1 = -\vec{v}_2$ ....(1)

$e = \frac{-2v_1}{12} \Rightarrow \frac{1}{3} = \frac{-2v_1}{12}$

$\Rightarrow v_1 = -2$ m/s and $v_2 = 2$ m/s

So, $|v_1| = |v_2| = 2$ m/s

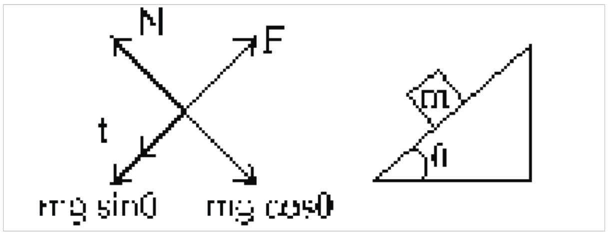
3) based on nuclear phenomenon.

15)

□ $J = nev_d = ne\mu E = m_1 e(\mu_e + \mu_h).E$

18) $F = v\frac{dm}{dt}$

20) In forward Biasing $V_P > V_n$

22) 

$F = f + mg\sin\theta = mg(\mu\cos\theta + \sin\theta)$

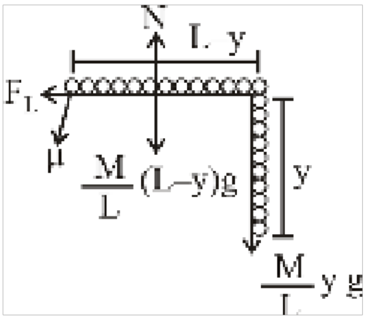
24) $a = v\frac{dv}{ds} = \frac{k^2}{2}$

$v = at = \frac{k^2t}{2}$

$W_{ALL} = \Delta K.E. = \frac{1}{2}m\left(\frac{k^2t}{2}\right)^2 = \frac{1}{8}mk^4t^2$

31) $T = \frac{M}{L}yg$ ...(1)

$F_L \geq T$ ...(2)



From (1) & (2)

$\mu\frac{M}{L}(l-y)g \geqslant \frac{M}{L}yg$

$\mu L \geqslant (\mu+1)y \Rightarrow y_{max} = \left[\frac{\mu_L}{1+\mu}\right]$

37) 100 – T = 0 ⇒ T = 100

T – 0.15 [m + 20] g = 0

100 – 1.5 [m + 20] = 0

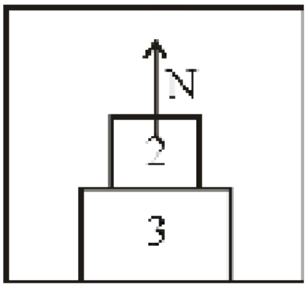
$\frac{15}{10}(m + 20) = 100$

$m + 20 = \frac{1000}{15}$

$m = \frac{1000-300}{15}$

$m = \frac{700}{15} = \frac{140}{3}$ Kg

38) Neutron is about 0.1 more massive than proton. But the unique thing about the neutron is that while it is heavy, it has no charge (it is neutral). This lack of charge gives it the ability to penetrate matter without interacting as quickly as the beta particles or alpha particles.

39) 

N = 2(g + a)

= 2 × 12

= 24

43) $V = \frac{4}{3}\pi\left(R_0A^{1/3}\right)^3 = \frac{4}{3}\pi R_0^3 \times A$

V' = 2V ⇒ A' = 2A = 120

⇒ Z = 50

48) $U = \frac{A}{r^3} - \frac{B}{r^2}$

$F = -\frac{dU}{dr} = 0$ for equilibrium

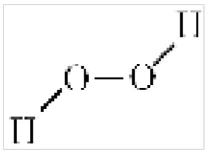
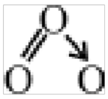
$-\frac{d}{dr}\left[\frac{A}{r^3} - \frac{B}{r^2}\right] = 0$

$-\frac{3A}{r^2} + \frac{2B}{r} = 0$

$\frac{3A}{r^2} = \frac{2B}{r} \Rightarrow r = \frac{3A}{2B}$

CHEMISTRY

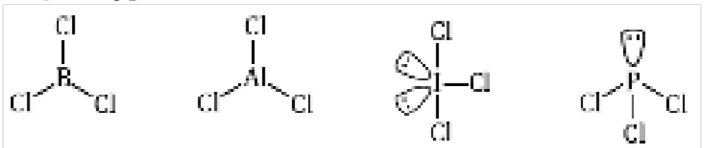
54) $\text{Thermal stability} \propto \frac{1}{\text{Bond length}}$

57)  >  > O = O (as per bond order)

64)

At this point in aqueous solution, the carboxyl group can lose a proton and amino group can accept a proton, giving rise to dipolar ion known as zwitter ion.

70) $AlCl_3$, BCl_3 → Hypovalent
ICl_3 → Hypervalent



73) NCERT Pg - 410/411

74) NCERT Pg. # 413 (Table 14.2)

75) Ncert class 12th, part-2, Article No: 10.1.2.1, 10.1.2.2, 10.1.4, Edition-2023-24.

80)

2p-2p axial overlapping.

87)

4 bond pair

2 lone pair
sp^3d^2,
Shape – Square Planner

93) Bond order $=\frac{1}{2}[N_b - N_a]$

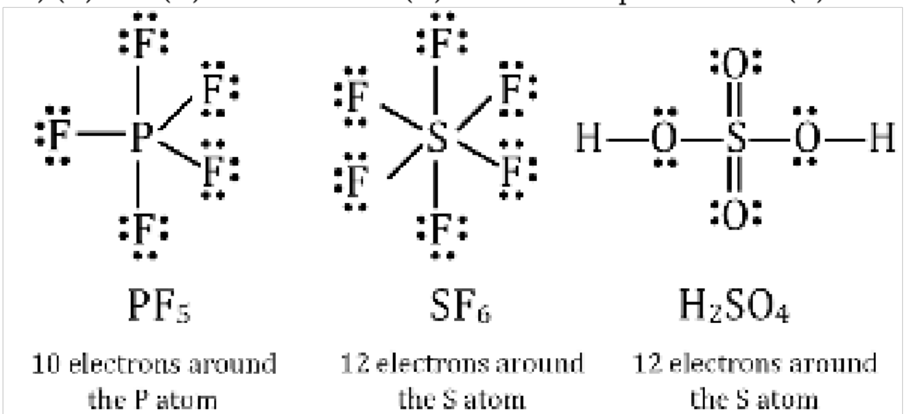
95) A positive bond order (i.e., $N_b > N_a$) means a stable molecule while a negative (i.e., $N_b<N_a$) or zero (i.e., $N_b = N_a$) bond order means an unstable molecule.

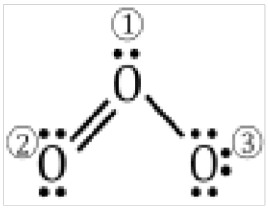
96)

NH_3 > NF_3 > BF_3
(1.47D) (0.23D) (0 D)

97) Obviously ionic bonds will be formed more easily between elements with comparatively low ionization enthalpies and elements with comparatively high negative value of electron gain enthalpy.

98) (A) and (R) is correct and (R) is correct explanation of (A).



99) 

1st Oxygen $6-2-\frac{1}{2}(6) = +1$

2nd Oxygen $6-4-\frac{1}{2}(4) = 0$

3rd Oxygen $6-6-\frac{1}{2}(2) = -1$

100) NF_5 [N belongs to 2nd period so it does not have d-orbitals]
BiI_5 [Bi is in +5 O.S → due to inert pair effect Bi^{+5} is not stable]

PbI_4 [Pb is in +4 O.S → due to inert pair effect Pb^{+4} is not stable]

BOTANY

105) NCERT Pg. # 05

108)

NCERT XI Pg. # 13

109)

NCERT (XIth) Pg. # 13

110) NCERT XI, Pg. # 16

112) NCERT (XI) Pg # 19

117)

NCERT XI Pg # 21

119) NCERT-XI ; Pg # 21 ; Fig 2.4 (a), Para 2.2.2

120) NCERT (XI) Pg # 24

121) NCERT (XI^{th}) Pg. # 23, 24

122) NCERT-XI Pg # 23

123)

NCERT - XI, Pg # 24, Para 03

125) NCERT Page # 27

126) NCERT Page # 27

136) NCERT Page # 4

141) NCERT-XI, Pg. # 13

142) NCERT XI Pg # 19, 20

143) NCERT XIth Pg.#21

145) NCERT (XI) Pg # 24, Para 2.3.4

146) NCERT Pg. # 27

147) NCERT XII Pg.# 244

148)

NCERT-XII, Pg. # 207, 208

149) B,C are correct

ZOOLOGY

151)

XII OLD NCERT Pg # 182

152) Some nations are developing laws to prevent unauthorised exploitation of their bio-resources and traditional knowledge. This is in response to the growing realisation of injustice and inadequate compensation and benefit sharing between developed and developing countries in the context of commercialisation of bio-resources.

153) NCERT Pg.#211

154) Statement I: Incorrect. While it is true that Basmati rice has a unique aroma and flavour and there are 27 documented varieties of Basmati grown in India, the statement is misleading as it suggests Basmati is exclusively grown in India, which is not the case. Basmati is also grown in other countries, such as Pakistan.
Statement II: Correct. In 1997, an American company did obtain patent rights on Basmati rice through the US Patent and Trademark Office, which allowed them to sell a 'new' variety of Basmati that was derived from Indian farmer’s varieties.

155) Statement I: Correct. The text describes the use of Agrobacterium vectors to introduce nematode-specific genes into the host plant, leading to the production of both sense and anti-

sense RNA. This results in the formation of dsRNA that triggers RNA interference (RNAi), silencing the specific mRNA of the nematode.
Statement II: Incorrect. The transgenic plant was able to protect itself from the nematode parasite by expressing specific interfering RNA, which silenced the mRNA of the nematode, preventing its survival in the host.

156)

NCERT XII Pg # 183, 184, 169

157) NCERT XII Pg. # 180

158)

NCERT Pg. No. 184

161) NCERT XI Pg # 288

168)

Lymph is modified blood that contains all contents of blood except RBC, platelets and certain proteins.

169)

NCERT XI Pg.No - 194

170) NCERT (XI) Pg. # 286

172)

NCERT XI Pg. # 280

173) NCERT XI (E) Pg # 288

178) NCERT (XI) Pg. # 272; para-5

181) NCERT Pg. No 185 & 186

183)

NCERT (XI) Pg. # 274

184)

NCERT-XI, Pg. # 187

185) NCERT (XI) Pg. # 271, 272

187) NCERT (XI) Pg. # 275; para-5

188) NCERT (XI) Pg. # 272; para-10

190) NCERT (XI) Pg. # 276; para-7

191) NCERT (XI) Pg. # 269; para-1

193)

NCERT-XI, Pg. # 185